

Ann prac2

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
# Input range
```

```
x = np.linspace(-10, 10, 400)
```

```
# -----
```

```
# Activation functions
```

```
# -----
```

```
def sigmoid(x):
```

```
    return 1 / (1 + np.exp(-x))
```

```
def tanh(x):
```

```
    return np.tanh(x)
```

```
def relu(x):
```

```
    return np.maximum(0, x)
```

```
def leaky_relu(x, alpha=0.1):
```

```
    return np.where(x > 0, x, alpha * x)
```

```
def softmax(x):
```

```
    exp_x = np.exp(x - np.max(x)) # stability fix
```

```
    return exp_x / np.sum(exp_x)
```

```
# -----
```

```
# Plot graphs
```

```
# -----
```

```
plt.figure(figsize=(12, 8))
```

```
# Sigmoid
plt.subplot(3, 2, 1)
plt.plot(x, sigmoid(x))
plt.title("Sigmoid Activation Function")
plt.grid(True)
```

```
# Tanh
plt.subplot(3, 2, 2)
plt.plot(x, tanh(x))
plt.title("Tanh Activation Function")
plt.grid(True)
```

```
# ReLU
plt.subplot(3, 2, 3)
plt.plot(x, relu(x))
plt.title("ReLU Activation Function")
plt.grid(True)
```

```
# Leaky ReLU
plt.subplot(3, 2, 4)
plt.plot(x, leaky_relu(x))
plt.title("Leaky ReLU Activation Function")
plt.grid(True)
```

```
# Softmax (illustrative)
plt.subplot(3, 2, 5)
plt.plot(x, softmax(x))
plt.title("Softmax Activation Function (1D)")
plt.grid(True)
```

```
# Main title
```

```
plt.suptitle("Activation Functions in Neural Networks", fontsize=14)
```

```
plt.tight_layout(rect=[0, 0.03, 1, 0.95])
```

```
plt.show()
```